

Contribution Report

Ensemble Methods: Boosting vs. Bagging
AI Academy – Machine Learning Final Project, Spring 2026

Team Sigmoid
July 2026

Team Members

Full Name	GitHub Username	Email
Emin Huseynli	ehuseynli96-ops	ehuseynli96@gmail.com
Ziyad Muradov	thedarkstringg	fgmz2014@gmail.com
Ismayil Yusifli	ismayilysfli	ismayilyusifli90@gmail.com

Individual Contributions

Member	Modules / Code	Report Sections	Experiments	Slides	Other
Emin Huseynli ehuseynli96-ops	src/trees/decision_tree.py src/boosting/ada_boost.py src/boosting/samme_r.py experiments/run_all.py experiments/bonus_samme_r.py	§2 Decision Tree §3 AdaBoost	Exp. 1, Exp. 2	Methods (Tree / AdaBoost)	Repo setup, .gitignore, initial scaffolding
Ziyad Muradov thedarkstringg	src/bagging/random_forest.py src/boosting/one_vs_rest_adaboost.py src/utils/preprocessing.py src/utils/datasets.py src/metrics/evaluation.py	§4 Random Forest §6 Experimental Setup	Exp. 3, Exp. 4, Exp. 5	Results (RF / CV tables)	requirements.txt, dataset acquisition (Kaggle CSV, verifying MNIST / Covertypes fetch)

Member	Modules / Code	Report Sections	Experiments	Slides	Other
Ismayil Yusifli ismayilysfli	src/unsupervised /pca.py src/unsupervised /kmeans.py src/unsupervised /dbscan.py experiments/*.py (1-7) experiments/bonus_tsne.py experiments/competition.py	§1 Introduction §7 Unsupervised Conclusion	Exp. 6, Exp. 7	Full deck assembly, intro / conclusion slides	Report LaTeX assembly, figures pipeline, contribution report compilation

Sign-off

All three members reviewed and approved the final codebase, report, and experimental results via GitHub pull request review.

By submitting this report, each undersigned member confirms that the contributions listed above accurately reflect their individual work on this project.

<i>Emin Huseynli</i>	<i>Ziyad Muradov</i>	<i>Ismayil Yusifli</i>
Emin Huseynli	Ziyad Muradov	Ismayil Yusifli
July 15, 2026	July 15, 2026	July 15, 2026